1) Create a DataFrame from the following data

```
= {
    'Name' = ['Alice', 'bob', 'charlie', 'David'],
    'Age' = [25, 30, 35, 40]
    'City': ['New york', 'Los Angels', 'Chicago', 'Houston']
```

Display dataframe:

Output:

```
df= pd. Dataframe (data)
print (df)
```

output:

	Name	Age	city
0	Alice	25	New york
1	Bob	30	Los Angeb
2	charlie	35	Chicago
3	David	40	Houston

2) Add a new column Salary with values [50000, 60000, 70000, 80000] to exist

Code:

```
df - Pd. Dataframe (data)
df['Salary'] =[50000, 60000, 70000, 80000]
```

output:

	Name	Age	City	Salary
0	Alice	25	New york	50000
1	bob	30	Los Angels	60000
2	charlie	35	chicago	70000
3	David	40	Houston	80000

3 Filtering Row. From the above Dataframe df Print only the rows when Age is greater than 28

Ans:

```
Python
Copy Edit
import pandas as pd
df - pd. Dataframe (data)
Filtered df = df[df['Age'] >28]
print [ filtered -df)
```

output

	Name	Age	city
1	Bob	30	Los Angelse
2	charlie	35	Chicago
3	david	40	Haston

4 update column vulve:

Increase the salary of all ppl by 10%

Code:

```
df=pd. Dataframe (data)
df['Salary'] = df ['salary'] *% 10
print(df)
```

	Name	Age	city	Salary
0	Alice	25	Newyork	55000.0
1	Bob	30	Los Angles	66000.0
2	charlie	35	chicago	77000.0
3	David	40	Houston	88000.0

ID
1
2
3
4
5
6
7
8
9
10
11
12

ID	Age	Height	weight	Income	Exp	Department	Rating	City	Salary
1	22	165	55	28000	1	IT	3.1		4xxxx
2	25	170	62	33000	2	HR	3.8		4x000
3	26	175	70	38000	4	IT	4.1		55000
4	NA	168	65	41000	5	Sales	4.8		6x000
5	35	NA	78	43000	7	IT	4.0		68000
6	42	180	NA	52000	10	Sales	4.7		[redacted]
7	48	172	82	NA	12	HR	3.3		73000
8	52	175	85	68000	15	IT	NA		71000
9	29	169	68	39000	5	IT	3.9		8x000
10	31	171	72	45000	6	Sales	4.2		91000
11	67	178	90	72000	20	HR	2.8		NA
12	75	187	105	85000	25	Sales	2.9		76000
									91000

```
import pandas as pd
df = pd.read_csv('Employee.csv')
print (df.isnull().sum())
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['city'] = df['city'].fillna(df['city'].mode()[0])
print (df)
```

5) Given a Dataframe with columns Department and Salary find the average salary per department

Code:

```
import pandas as pd
Data = {
    'Department': ['HR', 'IT', 'HR', 'IT', 'HR'],
    'Salary': [40000, 50000, 42000, 62000,
               50000]
}
df = pd.DataFrame(Data)
mean_Salary = df.groupby('Department')
              ['Salary'].mean()
Print(mean_Salary)
```

6) Replace all occurences of 'New York' in city with 'NYC'

Code:

```
import Pandas as pd
df['city'] = df['city'].replace('New York',
                                'NYC')
Print(df)
```

7) Drop the Age column from the Data frame

Code:

```
import Pandas as pd
df = df.drop(columns=['Age'])
Print(df)
```

8) Sort the Data fram

Code:

```
import pando
df - Sorted

Print
```

9) Check for mi

Code:

```
import
Print (
Print
```

10) Solved Bas

```
1 Plt.Plot(x
2 Plt.Scatte
3 Plt.bar(x
4 Plt.barh
5 Plt.hist
6 Plt.box Pl
7 Plt.Pie C
```

8) Sort the DataFrame by Salary in descending order.

Code:

```
import Pandas as Pd
df-Sorted = df.Sort-values(by='salary',
                           axending=False)
Print(df-Sorted)
```

9) Check for missing values in the Data Frame.

Code:

```
import Pandas as Pd
Print(df.isnull())
Print(df.isnull().sum())
```

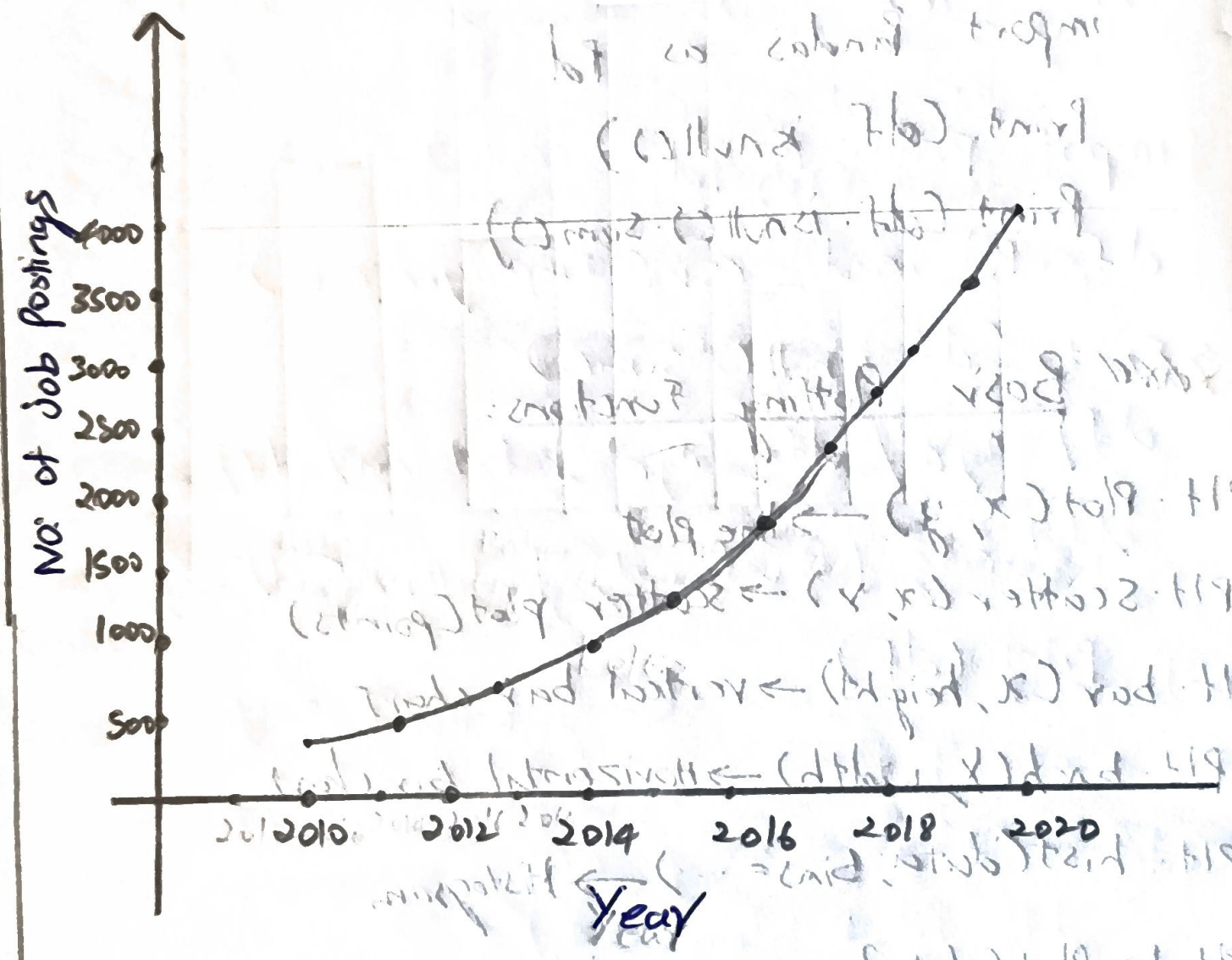
10) Solve: Basic Plotting Functions:

1 Plt.Plot(x, y) → line plot
2 Plt.Scatter(x, y) → scatter plot(points)
3 Plt.bar(x, height) → vertical bar chart
4 Plt.barh(x, width) → Hovizontal bar chart
5 Plt.hist(data, bins=...) → Histogram
6 Plt.boxPlot(data) → Box and whisker plot
7 Plt.Pie(data) → pie chart.

Sample Data Input

Year = 2010, 2021

Job posting = 150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200



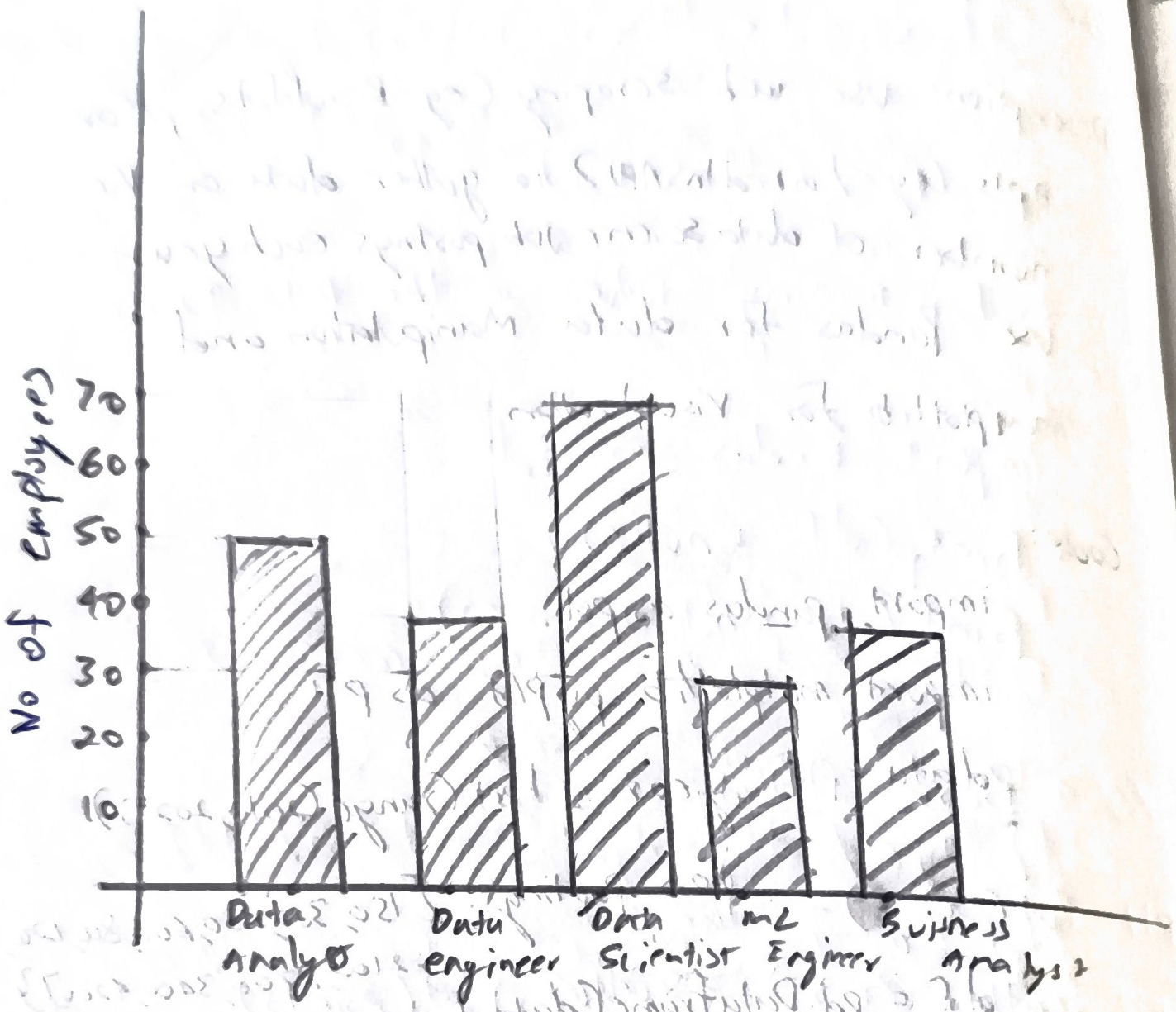
Exp No1

Analyse the trend of data science job postings over the last decade

Description: Use web scraping (eg: BeautifulSoup) or APIs (eg: LinkedIn API) to gather data on the number of data science job postings each year. Use Pandas for data manipulation and matplotlib for visualization.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'year': list(range(2010, 2021)),
        'Job Posting': [150, 300, 450, 600, 800, 1200, 1600
                        , 2100, 2700, 3400, 4200]}
df = pd.DataFrame(data)
plt.plot(df['year'], df['Job postings'], marker='o
plt.title('Trend of Data Science job posting')
plt.xlabel('year')
plt.ylabel('Number of Job postings')
plt.show()
```


No of employees
70
60
50
40
30
20
10
Data analyst
Data engineer
Data Scientist
ML Engineer
Business Analyst
Roles

Exp No 2 — Analyze and Visualize the distribution of various data Science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a data set of job postings and categorize them into different roles visualize the distribution using pie charts or bar plts

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML
        Engineer', 'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('count')
plt.show()
```

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst

Counter = 300, 500, 450, 200, 150

Exp No: 3

Write a Python Program to Demonstrate the three main types of Data Structured Unstructured and Semi Structural use a Pandas Data Frame for Structured data, a text string for Unstructured data and a dictionary for Semi-Structured data Print each type of data with clear labels.

Code:

```
import Pandas as Pd
Structured_data = Pd.DataFrame({
    'Name': ['Ann', 'Priya', 'Rahul'],
    'Age' : [20, 21, 19],
    'Department': ['CSE', 'ECE', 'IT']
    })
Unstructured_data = "Data Science is used
                     to Analyse and understand
                     complex data"
Semi-Structured data = {
    'id' : 101
    'Name' : 'Ann'
    'Skills' = ['Python', 'SQL', 'Machine Learning']
    }
Print ("Structured Data: ")
Print (Structured_data)
Print ("\n Unstructured Data: ")
```

```
Print(unstructured_data)
Print("\n Semi-structured Data:")
Print(Semi_structured_data)
```

Structured Data:

	Name	Age	Department
0	Arun	20	CSE
1	Priya	21	ECE
2	Rahul	19	IT

Unstructured Data:

Data Science is used to analyse and understa
Complex data

Semi-Structured Data:

{'id': 101, 'name': 'Arun', 'skills': ['Python', 'sql
, 'machine learning']

Exp no: 4

Write a Python Program using the cryptography in module to demonstrate Symmetric key encryption and decryption. Encrypt the text "Rajalakshmi engineering college" using a generated key, display the encrypted cipher text and then decrypt it back to the original text. Print the original, encrypted and decrypted data"

Code:

```
From cryptography.fernet import Fernet
key = fernet.generate_key
f = fernet(key)
token = f.encrypt(b"Rajalakshmi engineering
                   college")
token
f.decrypt(token)
key = Fernet.generateKey()
cipher_suite = Fernet(key)
original Data
~~plain_text~~ = b"Rajalakshmi Engineering college"
~~CIPH~~ Print("Original Data:", plain_text)
Print("Encrypted Data:", ciphertext)
Print("Decrypted Data:", decrypted_text)
Encrypted ~~Original~~ Data: b'gAAAAABgEALSYLTO9T
nS67gt99;40bVNN8Y"
Decrypted Data: Rajalakshmi Engineering
                College
```